

Hanyou Zheng

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Education

The Chinese University of Hong Kong

Ph.D. in Computer Science and Engineering

Advisor: Prof. Chi-Wing Fu

Aug 2025 – 2029 (expected)

Hong Kong SAR

Wuhan University

B.Eng. in Computer Science and Technology, Hongyi Honors Program

GPA: 3.8 / 4.0

Sep 2021 – Jun 2025

Wuhan, China

Research Interests

My research interests lie at the intersection of computer graphics and generative AI, with a focus on 3D content creation and computational design. I am also interested in AI agents, particularly in their applications to 2D/3D understanding, content creation, and computational design, along with broader interests in geometry processing and rendering.

Publications

- LEGO®-Maker: Autoregressive Image-Conditioned LEGO® Model Creation**
Jiahao Ge*, Mingjun Zhou*, **Hanyou Zheng**, Hao Xu[†], Chi-Wing Fu[†]
ACM Transactions on Graphics (Proceedings of SIGGRAPH Asia 2025), 44(6), 2025. [Project](#) / [DOI](#) / [Video](#)
* indicates equal contribution; [†] indicates corresponding authors.
- Space-Time Video Super-Resolution With Neural Operator**
Yuantong Zhang, **Hanyou Zheng**, Daiqin Yang, Zhenzhong Chen, Haichuan Ma, Wenpeng Ding
IEEE Transactions on Image Processing, 34:6742-6754, 2025. [DOI](#) / [arXiv](#) / [Code](#)
- NTIRE 2024 Restore Any Image Model (RAIM) in the Wild Challenge**
Jie Liang, Radu Timofte, ..., **Hanyou Zheng**, et al.
CVPR Workshops, pp. 6632-6640, 2024. [Paper](#)
- NTIRE 2023 Challenge on Efficient Super-Resolution: Methods and Results**
Yawei Li, Yulun Zhang, ..., **Hanyou Zheng**, et al.
CVPR Workshops, pp. 1922-1960, 2023. [Paper](#)

Research Experience

UNICUS Research (formerly Qianzhi Technology)

Jun 2024 – Present

Research Intern

- LEGO®-Maker: Autoregressive Image-Conditioned LEGO® Model Creation** (*ACM TOG, Proc. SIGGRAPH Asia 2025*). Built an image-conditioned autoregressive framework that uses LEGO tokenization to generate collision-free LEGO models from reference images with over 100 brick types; subsequently improved the system through synthetic training data, evaluation design, and generation-quality optimization.
- Graphics Engineering**. Developed tools for rendering-quality tuning, geometry processing, and large-scale LEGO asset processing and annotation.

Intelligent Information Processing Lab (IIP; now iAM), Wuhan University

Apr 2022 – Jun 2025

Undergraduate Research Student

Advisor: Prof. Zhenzhong Chen

- Space-Time Video Super-Resolution With Neural Operator** (*IEEE TIP, 2025*). Developed a neural-operator method for space-time video super-resolution that models motion estimation and compensation as a mapping between continuous function spaces, and uses Galerkin-type attention to handle large motions with global receptive fields and linear complexity.
- NTIRE 2024 Restore Any Image Model (RAIM) in the Wild Challenge** (*CVPR Workshops, 2024*). Developed DAIR, a degradation-aware image-restoration framework that models complex unknown degradations through per-pixel degradation-kernel estimation and representation injection, and uses generative priors to enhance texture details; ranked 2nd by objective score and 4th overall, receiving a Third-Class Award.
- NTIRE 2023 Challenge on Efficient Super-Resolution: Methods and Results** (*CVPR Workshops, 2023*). Proposed Diversified Local Feature Arch-Network for efficient image super-resolution under reconstruction-quality, model-complexity, and computational-cost constraints.

Selected Honors & Awards

Gold Medal, ICPC Asia Regional Contest (Shenyang)	Nov 2023
Gold Medal, ICPC Asia Regional Contest (Xi'an)	Oct 2023
Bronze Medal, ICPC East Asia Continent Final	Jan 2024
Third-Class Award, NTIRE 2024 RAIM Challenge (2nd by objective score, 4th overall)	Mar 2024

Teaching Experience

The Chinese University of Hong Kong	
Teaching Assistant · CSCI3260: Principles of Computer Graphics	Spring 2026 and Fall 2026
Teaching Assistant · CSCI1120: Introduction to Computing Using C++	Fall 2025
Wuhan University	
Teaching Assistant · Data Structures	Spring 2024